

120.green v1.FDW6 correction analysis

Analysis date: 2026-09-14

Conclusion

The v1.FDW6 filters are worth using with increased attenuation. They reduce a persistent 100-160 Hz excess at all five measured positions. Combined room group delay is nearly unchanged, broadband late energy generally falls, and no strong long pre-echo appears in the predicted corrected responses. The listener also prefers the result.

The BruteFIR shutdown is a separate gain-staging failure. Offline processing of the source track reproduces the logged peak within 0.01 dB.

Exact inputs

Only measurements in output/clean-fdw6/clean-fdw6.txts were used:

Position	Left	Right
Centre	L.0.txt	R.0.txt
1, 20 cm forward	L 120.green.1.txt	R 120.green.1.txt
2, approximately 20 cm back	L 120.green.2.txt	R 120.green.2.txt
3, 20 cm left	L 120.green.3.txt	R 120.green.3.txt
4, 20 cm right	L 120.green.4.txt	R 120.green.4.txt

Position 2's note says AHEAD , but its 4.387 m front-wall distance identifies it as the rear position. The label does not affect the data.

Applied filter	SHA-256
FLX-trimmed-48k.wav	9b1d092ff41b436a58bd0e9fef020c1493c8cbb951d2336a2e84017c7c7b6cc8
FRX-trimmed-48k.wav	1af03f6333967b872abd9302030e7126b20739d5d50113eb168132a01cfc787b

These match the deployed bundle's source WAVs. All installed RAW coefficient hashes match its manifest.

Method

Each complex response was multiplied by the corresponding exact filter. At each position:

```
before = (L + R) / 2
after  = (L * FLX + R * FRX) / 2
```

The shared acoustic timing reference preserves inter-channel timing. Common 8192-sample filter latency was removed, while relative phase was retained. Because measurements start near 15 Hz, time comparisons use identical tapered analysis windows. These are linear predictions, not fresh corrected sweeps.

Tonal correction across five positions

Coherent left-plus-right levels are referenced to each position's 500-2000 Hz mean.

Band	Before	After	Median change	Position range
20-80 Hz	+1.56 dB	+1.48 dB	-0.05 dB	-0.11 to +0.01 dB
80-100 Hz	+5.76 dB	+4.61 dB	-1.15 dB	-1.15 to -1.08 dB
100-160 Hz	+6.44 dB	+2.99 dB	-3.41 dB	-3.60 to -3.17 dB
160-225 Hz	+3.19 dB	+2.18 dB	-1.00 dB	-1.10 to -0.92 dB

The highest 100-160 Hz value falls by 3.41-4.83 dB at every position, median 4.52 dB. The filter therefore corrects a spatially persistent upper-bass excess rather than a centre-position accident. It intentionally changes little below 80 Hz.

Timing in room context

RMS group delay over 20-225 Hz, referenced to 300-500 Hz:

Position	Before	After	Change
Centre	27.18 ms	27.40 ms	+0.22 ms
1	22.93 ms	22.38 ms	-0.55 ms
2	33.08 ms	31.54 ms	-1.55 ms
3	30.11 ms	28.84 ms	-1.27 ms
4	25.79 ms	24.95 ms	-0.84 ms

Four positions improve slightly; the centre worsens by 0.22 ms. The median is effectively unchanged at 27.18 to 27.40 ms. The filter's few milliseconds of phase movement sit inside an existing 23-33 ms room response.

The broadband 20 Hz-15 kHz prediction shows the strongest energy earlier than 1 ms changing from roughly -59 to -64 dB before correction to -41 to -45 dB after. Earlier than 5 ms, corrected precursor level remains about -64 to -69 dB. Thus the phase correction adds a measurable near-main precursor but no strong long pre-echo across these positions.

Broadband energy	Median before	Median after	Across positions
Later than 20 ms	0.459%	0.411%	lower at all five
Later than 50 ms	0.062%	0.054%	lower at four, effectively unchanged at one
Later than 100 ms	0.0031%	0.0028%	lower at all five

These fractions are not reverberation times. Narrow bass windows have long symmetric impulses, and EQ changes their spectral weighting, so their raw pre/post energy must not automatically be labeled audible ringing.

Filter-only acceptance

Current revision 2 `dirc_acceptance.py` passes both source WAVs and installed 192 kHz coefficients. All nine estimator regression tests pass.

Check	Left	Right	Limit
Feature sharpness	Q 8.8	Q 3.4	Q <= 12
Group-delay excursion	+5.2 ms at 78.83 Hz	+7.0 ms at 95.95 Hz	magnitude <= 10 ms
Worst tail above control	+79 ms at 79 Hz	+80 ms at 100 Hz	<= 100 ms

This is a careful bass-resonance screen, not a perceptual certificate. Pre-peak energy does not enter its verdict, it uses median gate phase, and defaults stop at 200 Hz. Its 100 ms allowance was calibrated with FDW6 among the references.

Reproduced overload

The Qobuz bridge selected **Roxy Music - The Space Between** at 15:53:43. BruteFIR subsequently reported:

```
peak: 0/58/+4.69 1/66/+4.85
Safety limit exceeded on output (6.17 > 6.00). Aborting.
```

The 44.1 kHz, 16-bit source reaches about -0.004 dBFS during 85-87 seconds but does not hit integer rails there. SoXR resampling to 192 kHz exposes +0.162 dBFS left and +0.409 dBFS right. Convolution with the exact 192 kHz filters and configured 1.1 dB attenuation reaches **+6.179 dBFS right at 84.8268 seconds**, matching the shutdown within 0.01 dB.

The event follows from a near-full-scale master, intersample overshoot and phase-dependent peak growth. The minimum total attenuation for this event is about 7.28 dB. Eight dB leaves 0.72 dB; **10 dB total is the safer provisional setting**, leaving 2.72 dB. More recordings are needed for a universal bound.

Digital attenuation in the 64-bit floating-point, 32-bit-output chain does not alter phase or create harmonic ringing. Its cost is reduced level relative to the DAC noise floor.

Comparison with Dirac Live

Ordinary Dirac Live does not expose this build's user-controlled common low-frequency plus per-channel construction. That transparency is an advantage: each component can be inspected and tested independently.

Dirac Live Bass Control is related. Dirac documents a common low-frequency target, speaker-group-specific higher-frequency targets and cross-channel optimization, but its filter construction is automated and proprietary.

Dirac says its mixed-phase correction limits pre-ringing by correcting excess phase only when common across positions. It could produce less near-main precursor than an unconstrained one-position phase inverse. FDW6 does not use that kind of inverse: its magnitude divisors are minimum phase and its all-pass term corrects the known crossover. Five-position prediction shows low long precursor and stable combined timing.

There is no evidence that Dirac would produce a meaningfully better pre-echo result here. It might reduce the short precursor, or retain similar crossover correction because that behavior is common. Only a Dirac filter generated from the same measurements and tested identically could answer quantitatively.

References:

- [Dirac Live technical overview](#)
- [Dirac Live Bass Control filter design](#)

Final assessment

The intended correction is substantial and spatially consistent. Combined room delay is essentially unchanged, late broadband energy generally falls, and the new precursor is short. Filter acceptance passes, and listening favors the correction. The evidence supports continued use. The required remediation is gain staging: 1.1 dB attenuation is demonstrably insufficient.